

Shi Tang

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- Focus on learning systems allowing the rapid establishment of models with lower requirements on data and ML expertise.
- Familiar with metric-based few-shot learning, have worked on improving novel-class generalization of these methods.
- Currently interested in developing novel meta-learning methods better capable of handling the complexity of task space.

EDUCATION

Tsinghua University

Master of Software Engineering

Beijing, China

Sep. 2021 - Jun. 2024

- **GPA:** 3.73/4.0
- **Research:** few-shot learning and meta-learning, advised by [Prof. Guiming Luo](#)
- **Core courses:** Deep Learning (A), Automaton and Formal Logic (A), Data Stream System Modeling and Simulation (A-)

Dalian University of Technology

Bachelor of Software Engineering

Dalian, China

Sep. 2017 - Jun. 2021

- **GPA:** 3.81/5.0
- **Rank:** 3rd/84
- **Research:** depth estimation, advised by [Prof. Xinchen Ye](#)
- **Core courses:** Machine Learning (98), Signal Processing (95), Computer Vision (91), Data Structure and Algorithm (94)

PUBLICATIONS & PREPRINTS

Unleash the Power of Local Representations for Few-Shot Classification

Shi Tang, Guiming Luo, Xinchen Ye, Zhiyi Xia

Under review at Pattern Recognition

- Focus on the unexploited potential of local representations in improving novel-class generalization.
- Propose a novel pretraining paradigm for few-shot learning and an adaptive metric to handle various set combinations.
- Propose a UniCon KL-Divergence for distilling few-shot learning nets with both theoretical and empirical supports.
- **New state of the art** on five popular benchmarks, **outperforms transductive and cross-modal methods** in the fine-grained scenario.

UniEM-3M: A Universal Electron Micrograph Dataset for Microstructural Segmentation and Generation

Nan Wang, Zhiyi Xia, Yiming Li, Shi Tang, Zuxin Fan, et al.

Under review at ECCV

- Introduce UniEM3M, the first large-scale and multimodal EM dataset for instance-level understanding.
- Release a text-to-image diffusion model trained on the entire collection to serve as both a powerful data augmentation tool and a proxy for the complete data distribution.
- Responsible for developing a flow-based method to detect overlapping particles of TEM image.

Cross-Modality Depth Estimation via Unsupervised Stereo RGB-to-Infrared Translation

Shi Tang, Xinchen Ye, Fei Xue, Rui Xu

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023

- Propose to estimate depth in a cross-modal way to improve robustness to illumination and misleading textures.
- Propose a Fourier domain adaptation strategy and a multi-space warping regularization for synthesizing stereo IR images.
- Error reduction of **6.13%** and **5.10%** on D1-all against GWCNet and Monodepth, respectively.

PROJECTS

Parameter Residual Diffusion for Domain Adaptation

Jun. 2025 - present

- Generate model parameters for the target domain directly with **better performance and diversity**.
- Propose to generate the residual between source and target models for ease of generation.
- Incorporate low-rank decomposition of parameter matrices to increase the number of parameters that can be generated.

Meta-Learning for Cross-Subject Hand Gesture Recognition via Electromyography

Aug. 2022 - Sep. 2023

- Focus on cross-subject recognition that is seldom considered by previous research.
- Regard recognition for different subjects as tasks within one family, and propose to learn across them.
- Propose to divide EMG signals converted to Fourier domain into groups by frequency band and meta-learn the weights of different bands for different subjects.
- Accuracy improvement of **13.04%** against baseline in cross-subject setting.

Task-Specific Few-Shot Classification by Balancing Sample- and Class-Level Generalization

Jun. 2022 - Jun. 2023

- Aiming at few-shot classification tasks in real-world scenarios where new tasks may contain both base and novel classes.
- Propose to fuse features of normal and episodic pretraining weighted by a proposed Cross-Attention Module to balance sample- and class-level generalization task-specifically.
- Improvements of **16.30%**, **6.91%** and **1.46%** against normal pretraining, episodic pretraining and Meta-Baseline, respectively for accuracy on reconstructed miniImageNet under 1-shot setting.

PCB Defect Detection System based on Deep Learning

Sep. 2021 - present

- Propose a defect filtering technique based on metric learning, significantly reducing false positives.
- Improve the hierarchical network by avoiding modeling the joint probability distribution of defect categories and levels.
- Responsible for server maintenance.

Scene Depth Perception based on Binocular Infrared Camera

Sep. 2019 - Sep. 2020

- Extend the binocular depth estimation task from visible band to infrared band.
- Propose a novel domain adaption strategy and a feedback learning strategy to reduce the domain gap between different datasets.

Vehicle Mounted Multi-Band Stereo Vision Perception System

Aug. 2018 - Sep. 2019

- Responsible for improving the sub-pixel corner detection algorithm based on the checkerboard pattern, as well as the calibration and rectification of infrared cameras.
- Propose a novel infrared calibration board design scheme.
- **Won the bid for the Army Equipment Department's open tender project in July 2019.**

INTERNSHIP EXPERIENCE

DP Technology

Beijing, China

Research Intern, develop few-shot learning algorithms for segmenting electron microscope images.

Aug. 2024 - present

HKUST

Guangzhou, China

Research Intern, developed a meta-learning method for cross-subject hand gesture recognition.

Aug. 2022 - Sep. 2023

TEACHING EXPERIENCE

Automaton and Formal Logic

Beijing, China

Tsinghua University

Fall 2022

Teaching Assistant, lecturing mid-term test, as well as designing the final test.

Formal Language and Automaton

Tsinghua University

Teaching Assistant, lecturing mid-term test, as well as designing the final test.

Beijing, China
Spring 2022 & 2023

AWARDS

2023 **Tsinghua Academic Scholarship**

2022 **Guoshuang Scholarship**

2021 **Outstanding Graduate (top 5%)**

2020 **Academic Excellence Scholarship (top 5%)**

2019 **Third Prize in CUMCM**

2019 **Academic Excellence Scholarship (top 5%)**

2018 **Academic Excellence Scholarship (top 5%)**

2018 **Lingshui Scholarship (top 3%)**